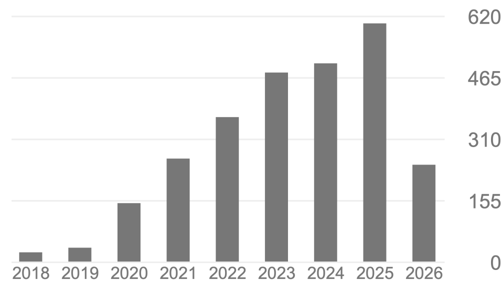
Sam Kriegman

Northwestern University, Tech A293
2145 Sheridan Road, Room B224, Evanston, IL 60208
samkriegman.com
www.xenobot.group
sam.kriegman@northwestern.edu

Curriculum Vitae

Key Academic Metrics

Citations:	2707
H-index:	19
Publications:	34
Invited Talks:	39
PhD Students:	8
Years since PhD:	6
Last updated:	*Jun 5, 2026*



Google Scholar

Research Statement

My research group explores the emergence of adaptive complexity in robots. Robots that not only compute and learn, but also design and build themselves and one another. Robots in other words, with **evolutionary agency.**

In pursuit of this, my research brings ideas, algorithms, and materials from biology to robotics and back. Our bioinspired approach is unique in that the entire agent—not just its brain (controller) but also its body (morphology)—is optimized from the ground up, tracing failures in behavior back (through simulation gradients) to errors or inefficiencies in particular parts of the body and brain. The resulting robots often embody adaptive structures and processes characteristic of organisms or their populations. But some of our robots are themselves organisms in a very real sense: they are composed entirely of biological cells.

Why? My students and I build robots for research, for amusement, and for both. We are driven by curiosity and interesting problems, rather than specific applications. By exploring analogies of evolution in robots we hope to make contact with the processes that produce novelty in nature. The robot thus becomes a scientific tool, a physical model that can make an abstract concept visceral. We also enjoy building prototypes, "concept robots" that demonstrate the feasibility of robots that repair, restructure, and reproduce themselves. Such capabilities would be as useful in a robot as they are in organisms.

Academic Positions

2022– **Assistant Professor**
Department of Computer Science (50%)
Department of Mechanical Engineering (25%)
Department of Chemical & Biological Engineering (25%)
McCormick School of Engineering, Northwestern University

2021–2022 **Postdoctoral Fellow**, Dept. of Biology, Tufts; Wyss Institute, Harvard. Advisor: Michael Levin.
2020–2021 **Postdoctoral Associate**, Dept. of Computer Science, University of Vermont. Advisor: Josh Bongard.

Education

2016–2020 **Ph.D.**, Computer Science, University of Vermont, USA. Advisor: Josh Bongard.

Dissertation: *Design for an Increasingly Protean Machine*.

2014–2016 **M.S.**, Statistics, University of Vermont, USA
2006–2010 **B.S.**, Applied Mathematics, Ohio University, USA

Funding

2025– **NSF CAREER** award, $576,083.
Differentiable Evolution: Efficient Automatic Design of Embodied Intelligence.

2024– **NSF FRR** award, $814,605. Kriegman portion: $413,909.
D Blackiston and S Kriegman
Modular biological robots with variable morphology.

2024– **Schmidt Sciences AI2050** Collaboration Fund, $120,000.

2023– **Templeton World Charity Foundation** award, $1,749,983. Kriegman portion: $286,600.
PI: J Foster; co-PIs: C Bergstrom, D Krakauer, S Kriegman, M Mitchell, R Rao.
Building Diverse Intelligences through Compositionality and Mechanism Design

2023–2026 **Schmidt Futures AI2050 Early Career Fellowship**, $300,000

2024–2025 **Berggruen Institute Fellowship**, $56,000

2023–2024 **CESR** Seed Grant, $125,000. Kriegman portion: $65,000.
S Kriegman and R Truby
Sustainable Design and Fabrication of Intelligent Robots

Honors and Awards

2025 CAREER Award, National Science Foundation

2024 Kavli Fellow, National Academy of Sciences

2023 Distinguished Early-Career Investigator Award, International Society for Artificial Life

2022 AI2050 Early Career Fellow, Schmidt Sciences
Outstanding Paper of 2021 Award, International Society for Artificial Life

2021 The Cozzarelli Prize, National Academy of Sciences
Outstanding Doctoral Dissertation Award, University of Vermont
Outstanding Paper of 2020 Award, International Society for Artificial Life
Altmetric Top 100, Altmetric

2020 Beazley Designs of the Year, The Design Museum
Top 10 Most Influential BioTech Projects, Project Management Institute
Computer Science Graduate Award, University of Vermont

Articles

13. C Yu, D Matthews, J Wang, J Gu, D Blackiston, M Rubenstein, S Kriegman (2026).
Agile legged locomotion in reconfigurable modular robots.
Proceedings of the National Academy of Sciences, 123(10), e2519129123.

12. D Matthews, A Spielberg, D Rus, S Kriegman, J Bongard (2023).
Efficient automatic design of robots.
Proceedings of the National Academy of Sciences, 120(41): e2305180120.

11. S Beaulieu, S Kriegman (2023).
Glamour muscles: why having a body is not what it means to be embodied.
ArXiv Preprint, arXiv:2307.08598

10. D Blackiston, S Kriegman, J Bongard, M Levin (2023).
Biological Robots: Perspectives on an Emerging Interdisciplinary Field.
Soft Robotics, 10.1089/soro.2022.0142.

9. D Kudithipudi, M Aguilar-Simon, J Babb, M Bazhenov, D Blackiston, J Bongard, AP Brna, S Chakravarthi Raja, N Cheney, J Clune, A Daram, S Fusi, P Helfer, L Kay, N Ketz, Z Kira, S Kolouri, JL Krichmar, S Kriegman, M Levin, S Madireddy, S Manicka, A Marjaninejad, B McNaughton, R Miikkulainen, Z Navratilova, T Pandit, A Parker, PK Pilly, S Risi, TJ Sejnowski, A Soltoggio, N Soures, AS Tolias, D Urbina-Melendez, FJ Valero-Cueva, GM van de Ven, JT Vogelstein, F Wang, R Weiss, A Yanguas-Gil, X Zou, H Siegelmann (2022).
Biological underpinnings of lifelong learning machines.
Nature Machine Intelligence, 4(3): 196-210.

8. S Kriegman, D Blackiston, M Levin, J Bongard (2021).
Kinematic self-replication in reconfigurable organisms.
Proceedings of the National Academy of Sciences, 118(49): e2112672118.

7. D Blackiston, E Lederer, S Kriegman, S Garnier, J Bongard, M Levin (2021).
A cellular platform for the development of synthetic living machines.
Science Robotics, 6(52): eabf1571.

6. D Shah, J Powers, L Tilton, S Kriegman, J Bongard, R Kramer-Bottiglio (2021).
A soft robot that adapts to environments through shape change.
Nature Machine Intelligence, 3(10): 51-59.

5. D Shah, B Yang, S Kriegman, M Levin, J Bongard, R Kramer-Bottiglio (2020).
Shape Changing Robots: Bioinspiration, Simulation, and Physical Realization.
Advanced Materials, 33(19): 2002882.

4. S Kriegman, D Blackiston, M Levin, J Bongard (2020).
A scalable pipeline for designing reconfigurable organisms.
Proceedings of the National Academy of Sciences, 117(4): 1853-1859.
(*A perspective article on this work by P. Ball can be found here.*)

3. S Kriegman (2019).
Why virtual creatures matter.
Nature Machine Intelligence, 1(10): 492.

2. S Kriegman, N Cheney, J Bongard (2018).
How morphological development can guide evolution.
Scientific Reports, 8(1): 13934.

1. F Corucci, N Cheney, S Kriegman, J Bongard, C Laschi (2017).
Evolutionary developmental soft robotics as a framework to study intelligence and adaptive behavior.
Frontiers in Robotics and AI, 4: 34.

Peer-reviewed Conference Publications

21. D Matthews, S Kriegman (2026).
Zero-shot robot design with residual physics.
Artificial Life Conference (ALife).

20. Z Guo, M Li, S Zhang, S Kriegman (*in review*).
Creating manufacturable blueprints for coarse-grained virtual robots.
Intl. Conf. on Intelligent Robots and Systems (IROS).

19. J Wang, C Yu, D Matthews, E Alexander, S Kriegman, M Rubenstein (2026).
Computational Design of a Low-Visibility UAV Using a Human-Aligned Perceptual Metric.
Robotics: Science and Systems (RSS).

18. Y Wang, M Li, Z Guo, S Kriegman (2026).
ECo-MoE: Embodiment-Conditioned Mixture of Experts Increases the Evolvability of Robots.
Intl. Conference on Machine Learning (ICML).

17. C Yu, S Kriegman (2026).
Robots that redesign themselves through kinematic self-destruction.
Intl. Conference on Robotics and Automation (ICRA).

16. L Strgar, S Kriegman (2026).
Accelerated co-design of robots through morphological pretraining.
Intl. Conference on Learning Representations (ICLR).

15. M Li, L Kong, S Kriegman (2025).
Generating Freeform Endoskeletal Robots.
Intl. Conference on Learning Representations (ICLR).
(Spotlight; top 1.4% of 11.6K submissions; top rated paper in subject area.)

14. L Strgar, D Matthews, T Hummer, S Kriegman (2024).
Evolution and learning in differentiable robots.
Robotics: Science and Systems (RSS), 10.15607/RSS.2024.XX.100.

13. T Hummer, S Kriegman (2024).
A non-cubic space-filling modular robot.
Intl. Conference on Robotics and Automation (ICRA), 2624-2631, 10.1109/ICRA57147.2024.10611176.

12. M Li, D Matthews, S Kriegman (2024).
Reinforcement learning for freeform robot design.
Intl. Conference on Robotics and Automation (ICRA), 8799-8806, 10.1109/ICRA57147.2024.10610048.

11. S Kriegman, A-M Nasab, D Blackiston, H Steele, M Levin, R Kramer-Bottiglio, J Bongard (2021).
Scale invariant robot behavior with fractals.
Robotics: Science and Systems (RSS), 10.15607/RSS.2021.XVII.059

10. J Powers, R Grindle, S Kriegman, L Frati, N Cheney, J Bongard (2020).
Morphology dictates learnability in neural controllers.
Artificial Life Conference (ALife), 52-59.

9. S Kriegman, A-M Nasab, D Shah, H Steele, G Branin, M Levin, J Bongard, R Kramer-Bottiglio (2020).
Scalable sim-to-real transfer of soft robot designs.
Intl. Conference on Soft Robotics (RoboSoft), 359-366, 10.1109/RoboSoft48309.2020.9116004.

8. D Matthews, S Kriegman, C Cappelle, J Bongard (2019).
Word2vec to behavior: morphology facilitates the grounding of language in machines.
Intl. Conf. on Intelligent Robots and Systems (IROS), 4153-4160, 10.1109/IROS40897.2019.8967639.

7. S Kriegman, S Walker, D Shah, M Levin, R Kramer-Bottiglio, J Bongard (2019).
Automated shapeshifting for function recovery in damaged robots.
Robotics: Science and Systems (RSS), 10.15607/RSS.2019.XV.028
(*A perspective article on this work by H. Hauser can be found here.*)

6. S Beaulieu, S Kriegman, J Bongard (2018).
Combating catastrophic forgetting with developmental compression.
Genetic and Evolutionary Computation Conference (GECCO), 386-393, 10.1145/3205455.3205615.

5. S Kriegman, N Cheney, F Corucci, J Bongard (2018).
Interoceptive robustness through environment-mediated morphological development.
Genetic and Evolutionary Computation Conference (GECCO), 109-116, 10.1145/3205455.3205529.

4. J Powers, S Kriegman, J Bongard (2018).
The effects of morphology and fitness on catastrophic interference.
Artificial Life Conference (ALife), 606-613.

3. S Kriegman, C Cappelle, F Corucci, A Bernatskiy, N Cheney, J Bongard (2017).
Simulating the evolution of soft and rigid-body robots.
Genetic and Evolutionary Computation Conference (GECCO), 1117-1120, 10.1145/3067695.3082051.

2. S Kriegman, N Cheney, F Corucci, J Bongard (2017).
A minimal developmental model can increase evolvability in soft robots.
Genetic and Evolutionary Computation Conference (GECCO), 131-138, 10.1145/3071178.3071296.

1. S Kriegman, M Szubert, J Bongard, C Skalka (2016).
Evolving spatially aggregated features from satellite imagery for regional modeling.
Parallel Problem Solving from Nature (PPSN), 707-716.
(*Best Paper Award Finalist.*)

Patents

2025 "Modular legged machines". Provisional.
2025 "Three-dimensional construction blocks with configuration-invariant attachment". Prov. No. 63/618,628
2024 "Efficient automatic design of physical machines with moving parts". US20240281571A1
2022 "Engineered multicellular ciliated organisms and the kinematic self-replication thereof". US20220220437A1
2021 "Engineered multicellular organisms". US20230235296A1

Service

2025– Area Chair, Robotics: Science and Systems (RSS).
2023–25 Program committee member, Genetic and Evolutionary Computation Conference (GECCO).
2022–24 Program committee member, Conference on Artificial Life (ALIFE).
2021–22 Co-organizer, CD-SoRo (computational design of soft robots) workshop, IROS conference. Kyoto, Japan.
2019–22 Co-developer, Voxcraft: a low-cost, open-source soft robot kit for ages 12+
2017–22 Co-organizer, Virtual Creatures Competition and Workshop.

Membership

2026– American Society of Mechanical Engineers (ASME)

2022–	International Society of Artificial Life (ISAL)
2022–	Association of Computing Machinery (ACM)
2022–	Institute of Electrical and Electronics Engineers (IEEE)

Reviewer

Science, *Nature*, *Nature Communications*, *Nature Machine Intelligence*, *Artificial Life*, *Soft Robotics*, *Transactions on Robotics*, *Robotics and Automation Magazine*, and others...

Internal

2026–	Embodied AI working group, McCormick
2026–	Biohybrid systems working group, McCormick
2026–	DTC working group, McCormick
May 2026	CAREER Workshop Panel, McCormick
2026–	Mentoring a research assistant professor (non-collaborator) for the Dept of Comp Sci
2024–	Faculty Search Committee for an "Embodied AI" hire, Depts of Comp Sci and Mech Eng
2022–25	Graduate Studies Committee, Dept of Mech Eng
2022–24	PhD Admissions Committee, Dept of Comp Sci

Teaching

Fall	Evolutionary Computation (Comp Sci 496).
Winter	ALife (Artificial Life; Comp Sci/Mech Eng/Chem Eng 302).
Spring	DTC (Design Thinking & Communication; Dsgn 106).

Advising

Staff

2022–2024	David Matthews: Differentiable evolution.

PhD's

2025–	Yibin Wang: Genetic representations (Comp Sci).
2025–	Daniel Shamsoddini: tbd (Comp Sci).
2024–	Yuchen Cao: tbd (Comp Sci).
2024–	David Matthews: Residual neural physics (Comp Sci). NSF GRFP recipient.
2024–	Zihan Guo: Sim-to-real automation (Mech Eng).
2023–	Luke Strgar: Differentiable robots (Comp Sci).
2023–	Chen Yu: Modularity (Comp Sci).
2023–	Muhan Li: Simulation (Comp Sci).

Masters

2025–	Xingyu Qiu (Elec & Comp Eng).
2025–	Hongyuan Xu (Comp Sci).
2025–	Shuzhe Zhang (Elec Eng).
2023–25	Lingji Kong (Comp Sci).
2023–24	Isabel Zhong (Biomedical Eng & Comp Sci).
2022–23	Tyler Hummer (Mech Eng).

Undergrads

2025–	Daniel Kıllıoğlu (Mech Eng).
2025–	Ben Li (Comp Sci).
2023–25	Daniel Shamsoddini (Comp Sci).

Invited Talks

Oct 2026 “Evolutionary Robotics”. Alumnae of Northwestern University Continuing Education Program.
Sept 2026 “Evolving robotic matter and metamachines”. IROS Conf. workshop. Pittsburgh, PA.
Jun 2026 “Fantastic machines and where to find them”. Midwest Robotics Workshop, TTIC, Chicago.
Apr 2026 “Explorations in Morphospace”. RSS AC Meeting, Columbia University.
Apr 2026 “Whole body engineering”. Wild Cat Days.
Apr 2026 “Possible Machines: Explorations in latent space”. ECE Seminar Series.
Mar 2026 “Evolving diverse intelligences”. Santa Fe Institute, New Mexico.
Mar 2026 *Evolutionary Robotics Masterclass*. Embodied Intelligence Conference.
Jan 2026 “A body is not all you need”. Mini Workshop on Generalist Robots & Embodied Intelligence.

Oct 2025 “Evolution, the impersonal engineer”. McCormick Advisory Council.
May 2025 “Replaying the movie of evolution”. Berggruen Institute.
Mar 2025 “From AI to ALife”. University of Wisconsin, Madison.

Oct 2024 “From AI to ALife”. Schmidt AI in Science Seminar Series, University of Chicago.
Mar 2024 “From Artificial Intelligence to Artificial Life”. NAS Kavli Frontiers of Science Symposium, Irvine, CA.
Mar 2024 “From Artificial Intelligence to Artificial Life”. NASEM Distinctive Voices Lecture, Irvine, CA.
Feb 2024 “Differentiable robot design”. Purdue University.

July, 2023 “From artificial intelligence to artificial life”. The American Academy of Arts and Sciences.
Apr, 2023 “AI-generated organisms”. Illinois Institute of Technology.
Mar, 2023 *Animal Robot* Screening and Discussion. AAAS Annual Meeting, Washington, DC.

Dec, 2022 *Berggruen Institute x Lucy McRae Salon*. Honor Fraser Gallery, Los Angeles.
Oct, 2022 “Selection, the impersonal engineer”. AI-Driven Labs Workshop, Argonne National Laboratory.
Sept, 2022 “AI-generated organisms”. New Faculty Invited Lecture, Northwestern ChBE Retreat.
May, 2022 “Everything I wish I’d known about the academic job market.”. University of Vermont.
Apr, 2022 “Simulating xenobots and xenohybrid machines.”. Workshop on software for soft robotics research.
Apr, 2022 “Sim2real for biological robots”. Workshop on soft robot design optimization, IEEE RoboSoft Conf.
Mar, 2022 “From Biology to Bots and Back”. Luddy School, Indiana University.
Mar, 2022 “From Biology to Bots and Back”. CS Colloquium, Northwestern University.
Feb, 2022 “Computer-designed organisms”. *Leonardo Art Science Evening Rendezvouz*, Stanford University.
Feb, 2022 “From Biology to Bots and Back”. MIT.
Feb, 2022 “Fractal robots”. Evolutionary and Learning Machines Group, Vrije Universiteit Amsterdam.
Feb, 2022 “From Biology to Bots and Back”. New York University.
Jan, 2022 “From Biology to Bots and Back”. Vanderbilt University.

Sept, 2021 “AutoCAD for XenoBOT”. Autodesk.
July, 2021 “Evolutionary robotics in a nutshell”. ISAL Summer School.
July, 2021 “Sim2Life: AI-generated biological constructs”. Cross Roads.
Mar, 2021 “Protean machines”. The Creative AI Lab, IT University of Copenhagen.
Mar, 2021 “Living robots”. *The Int’l Workshop on Embodied Intelligence*.
Mar, 2021 “How to evolve your robot”. Guest lecture, Introduction to Soft Robotics, Yale University.

Oct, 2020 “Living deepfakes”. Guest lecture for the MIT Media Lab’s Deepfakes course (MAS.S60).
Apr, 2020 “Computer designed organisms”. *Artificial Life Virtual Seminar Series*.
Feb, 2020 “Living robots for biomedicine”. Biomedical Engineering Society, University of Vermont.

Selected Media Coverage

Mar, 2026 “**A robot that can evolve**”. ***BBC Newshour*** (live, on air)
Mar, 2026 “AI designed ‘metamachines’ keep moving after damage”. *Reuters*
Nov, 2023 “**For first time, researchers use artificial intelligence to build a robot**”. ***CBS Evening News***
Oct, 2023 “**Generation AI: If you want something done**”. ***Reuters***

Oct, 2023	"AI designs new robot from scratch in seconds". *Reuters*
Oct, 2023	"AI creates a robot from scratch in seconds at Northwestern University". *CBS Morning News Chicago* (live)
Mar, 2023	"Animal Robot" documentary. *Scientific American and the Howard Hughes Medical Institute*
Sept, 2022	"Xenobot". *Dictionary.com*
Jan, 2022	**"Scientists create 'robots' that are capable of reproduction (with Jericka Duncan)". *CBS Evening News***
Dec, 2021	"Scientists Create 'Living Machines' With Algorithms, Frog Cells". *Bloomberg Businessweek*
Dec, 2021	**"It's not science fiction. Scientists have really made robots that reproduce". *NPR Weekend Edition***
Dec, 2021	"Living robots made in a lab have found a new way to self-replicate, researchers say". *NPR*
Nov, 2021	"These living robots made of frog cells can now reproduce, study says". *Washington Post*

Hundreds of additional articles appeared in the global press following our third xenobots paper.

Jun, 2021	"Biological Robots May Soon Build You a Better Heart". *Bloomberg Moonshot*
Dec, 2020	"The big scientific breakthroughs of 2020". *The Week*
Dec, 2020	"The 10 Most Spectacular Scientific Advances of 2020". *La Razón (Spain)*
Dec, 2020	"Part Robot, Part Frog: Xenobots Are the First Robots Made From Living Cells". *Discover Magazine*
Apr, 2020	"Meet the Xenobots: Virtual Creatures Brought to Life". *New York Times*
Apr, 2020	"What if, Instead of the Internet, We Had Xenobots? ". *New York Times*
Feb, 2020	"Giant Moon rocket, living robots and quantum computer – January's best science images". *Nature*
Feb, 2020	"Tiny machines made from the stem cells of frogs". *The Intelligence (Economist Radio)*
Jan, 2020	"A research team builds robots from living cells". *The Economist*
Jan, 2020	"The religious, moral, and ethical implications of Xenobots". *BBC Radio 4 Sunday*
Jan, 2020	"Scientists use stem cells from frogs to build first living robots". *The Guardian*
Jan, 2020	"Xenobot: how did earth's newest lifeforms get their name? ". *The Guardian*
Jan, 2020	"'Robots vivientes' hechos a partir de tejido de ranas, llamados Xenobots". *Noticieros Televisa*

Hundreds of additional articles appeared in the global press following our announcement of Xenobots.